

BOOTCAMP ON

AGENT-BASED MODELING

Impact Of Building Forms On Visual Perception
in urban environments
using Vision-Tracking Algorithm

SEPTEMBER 2024

Morphotect



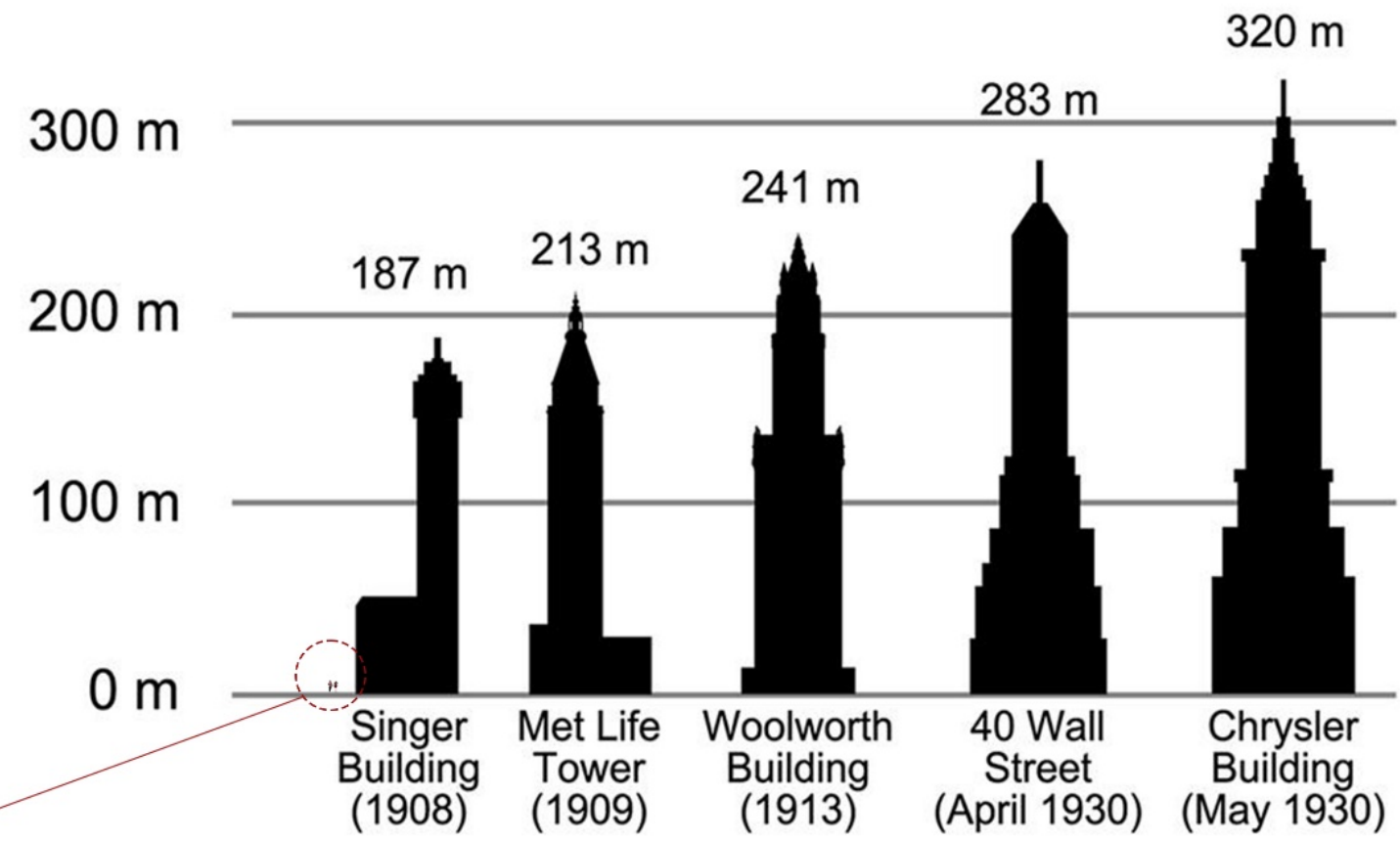
Elnaz Ghafourian

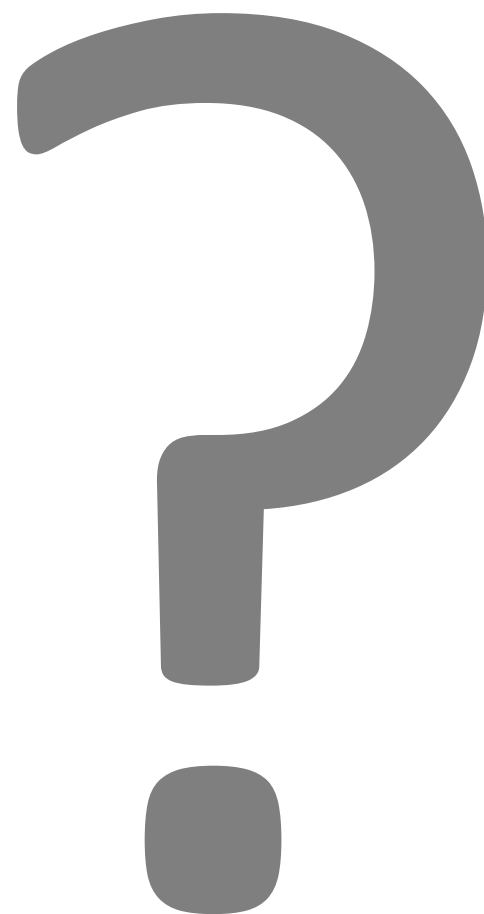
Fatemeh Ardeshir

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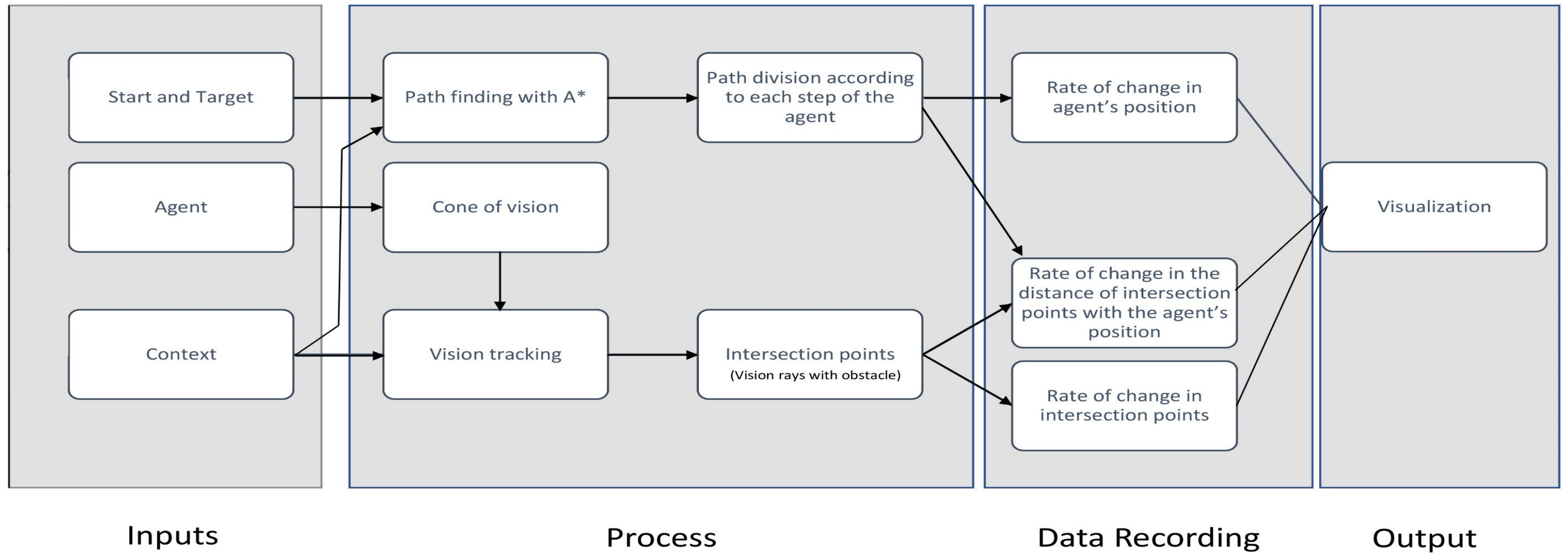
Question:

What is the impact of building forms on agent's visual perception in urban environments?

Building Forms:

- **Curved Forms**
- **Paralleled to the movement Path**
- **Diagonal to the movement Path**

Problem- Solving Proposal



Calculation Method

Vision Tracking Metrics:

- **Agent's Position**
- **Fixation Count**
- **Fixation Duration**



Calculation Method:

- **Angles of Tangent vectors of the movement path and the path**
- **Count of Intersection Points**
- **Rate of Change in Agent's Distance to Intersection Points**

Analysis Method

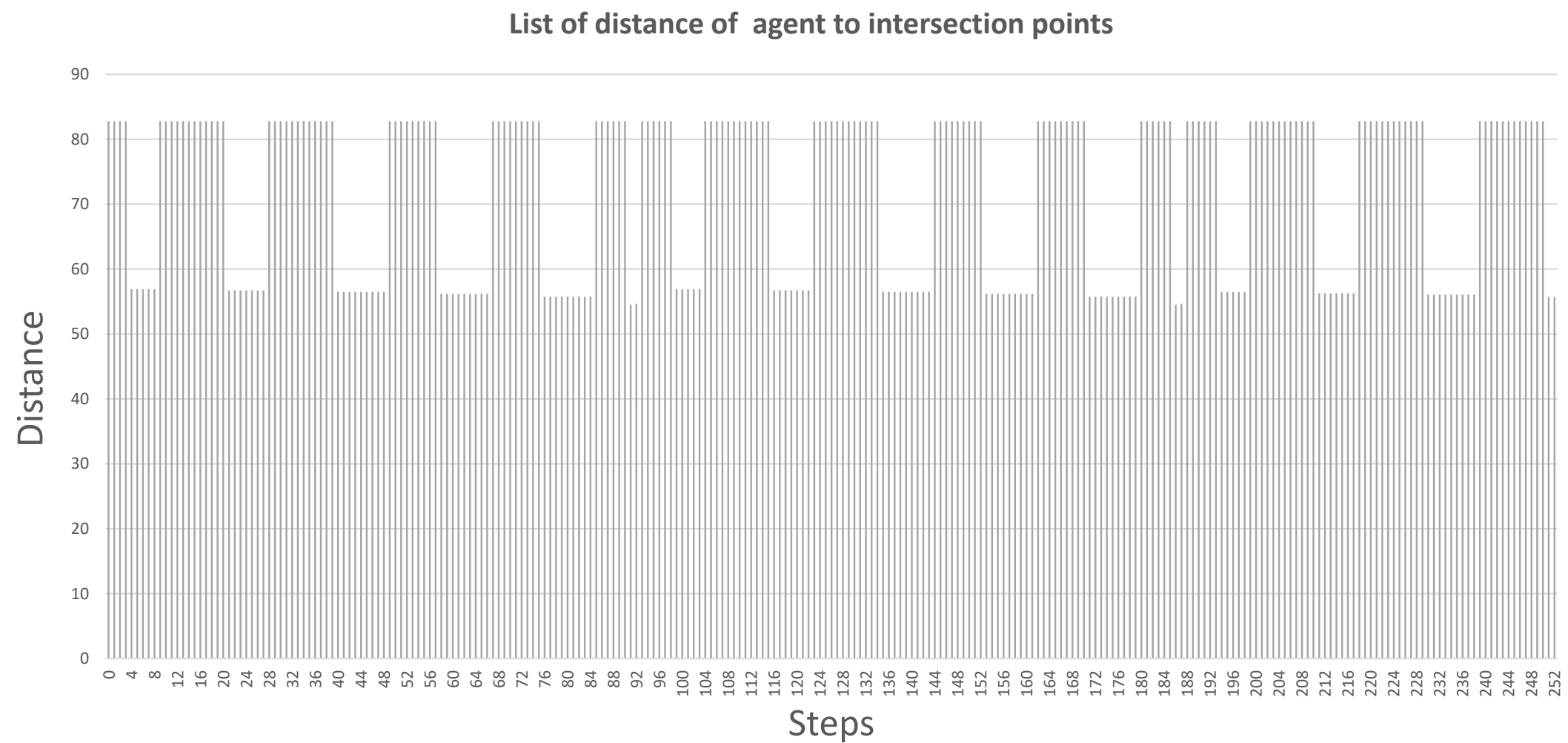
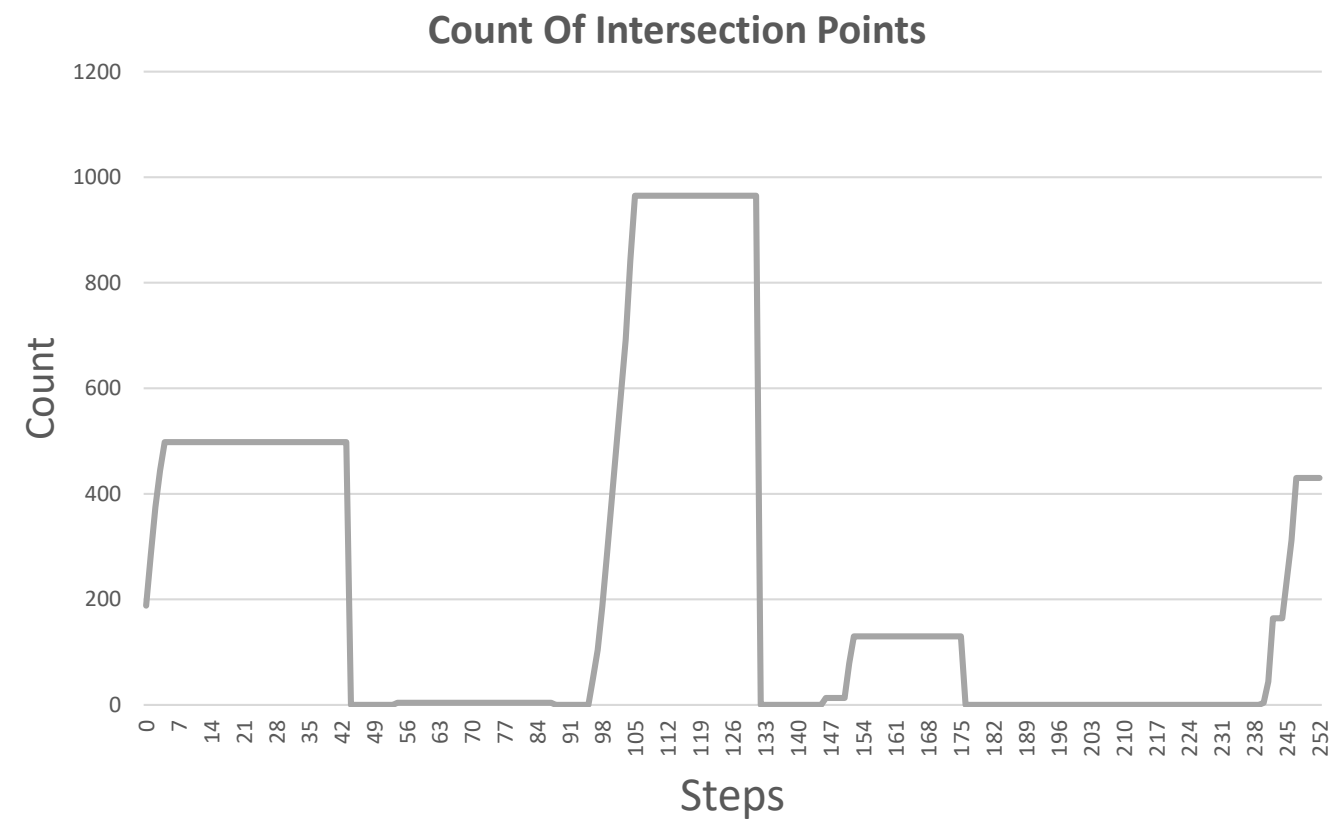
Calculation Method:

- **Angles of Tangent vectors of the movement path and the path**
- **Count of Intersection Points**
- **Rate of Change in Agent's Distance to Intersection Points**



Analysis Method:

- **Changes of Agent's Vision**
- **Visibility Degree**
- **Agent's Opportunity For Seeing the Urban Environment**



Rate Of Change in Distance of Agent to the Intersection Points

